Characterising and Controlling Complex Quantum Processes with Classical Memory

arXiv:2307.11905

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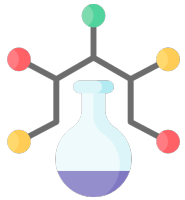
When present but **uncontrollable**, it manifests as **non-Markovian noise**…

WEATHER PATTERNS

FINANCIAL MARKETS



BIOCHEMICAL PROCESSES

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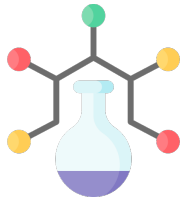

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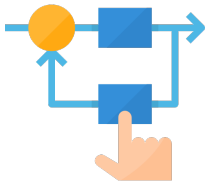
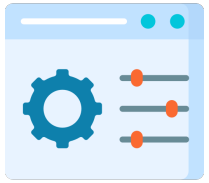
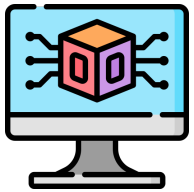
…Conversely, if **controllable**, it can be a **powerful resource** for various tasks of interest



FEEDBACK PROTOCOLS



OPTIMAL CONTROL



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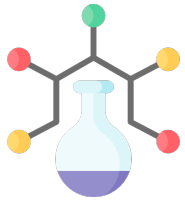

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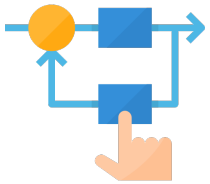

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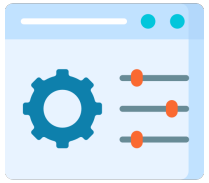

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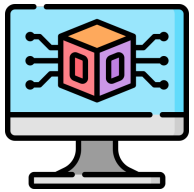

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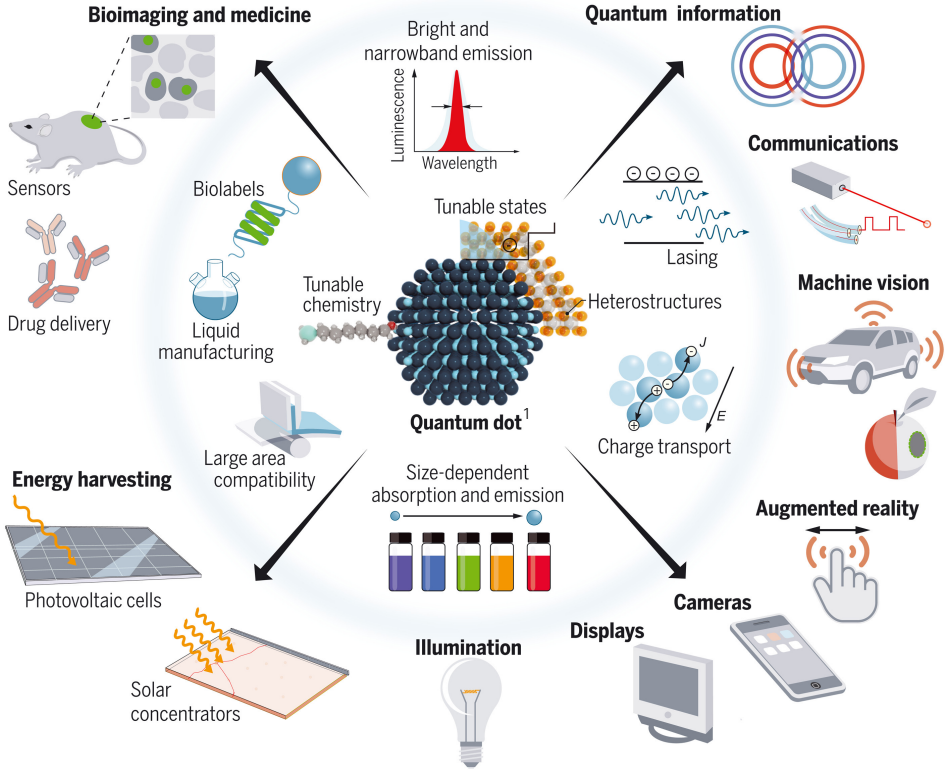

[1] F. Pelayo García de Arquer, D. V. Talapin, V. I. Klimov, Y. Arakawa, M. Bayer, E. H. Sargent, Science **373**, 6555 (2021)

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Memory effects arise from / are transmitted via interactions between the **system** and its **environment**

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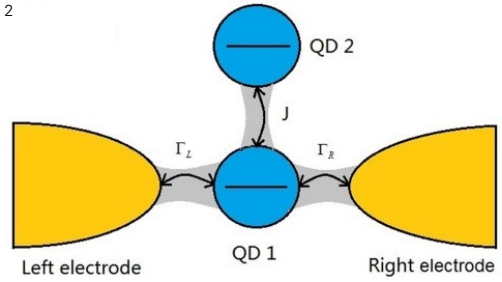

- Dynamics of a superconducting quantum dot between two electrodes
- As it evolves, information about its state influences environment, e.g. state of electrodes and neighbouring dots, inaccessible degrees of freedom, etc.
- In turn, these environmental features influence the future behaviour of the dot
- Environment acts as a memory carrier; will use the terms interchangeably

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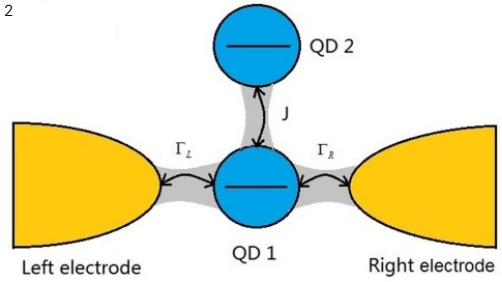

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Multi-time correlations are crucial to characterise, understand, and exploit

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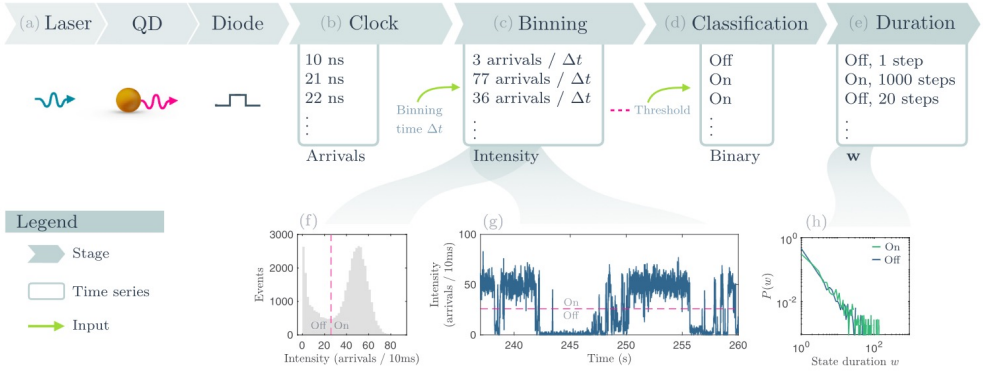

- Non-trivial memory effects in the temporal correlations of a blinking quantum dot
- Time between blinks follow a power-law distribution
- Such effects are inexplicable through transition distribution alone
- Multi-time effects play a significant role

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[3] R. N. Muñoz, L. Frazer, G. Yuan, P. Mulvaney, F. A. Pollock, K. Modi, Phys. Rev. E **106**, 014127 (2022)

Quantum processes with classical memory promise *near-term application*

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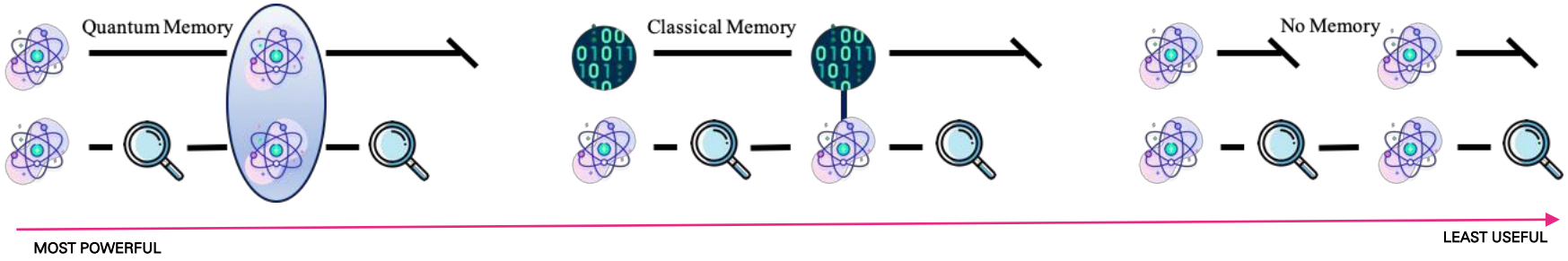

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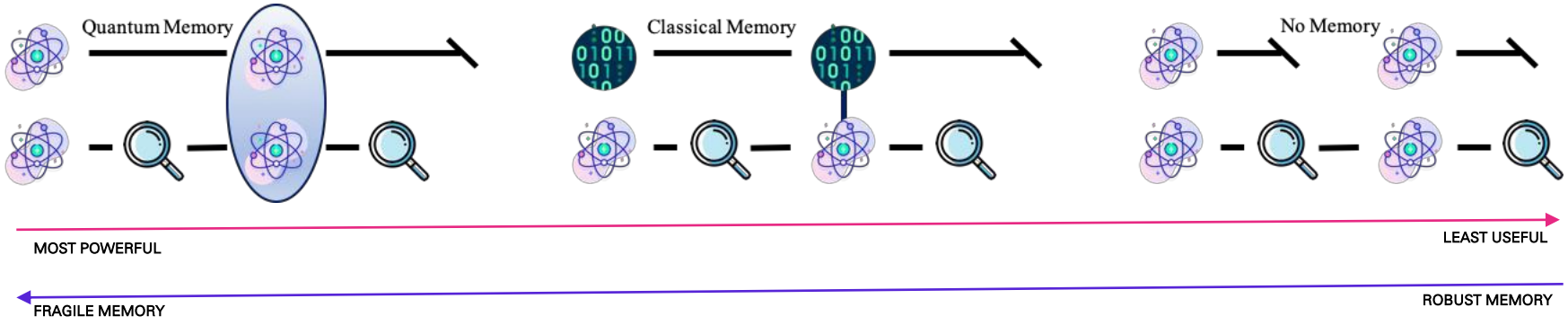

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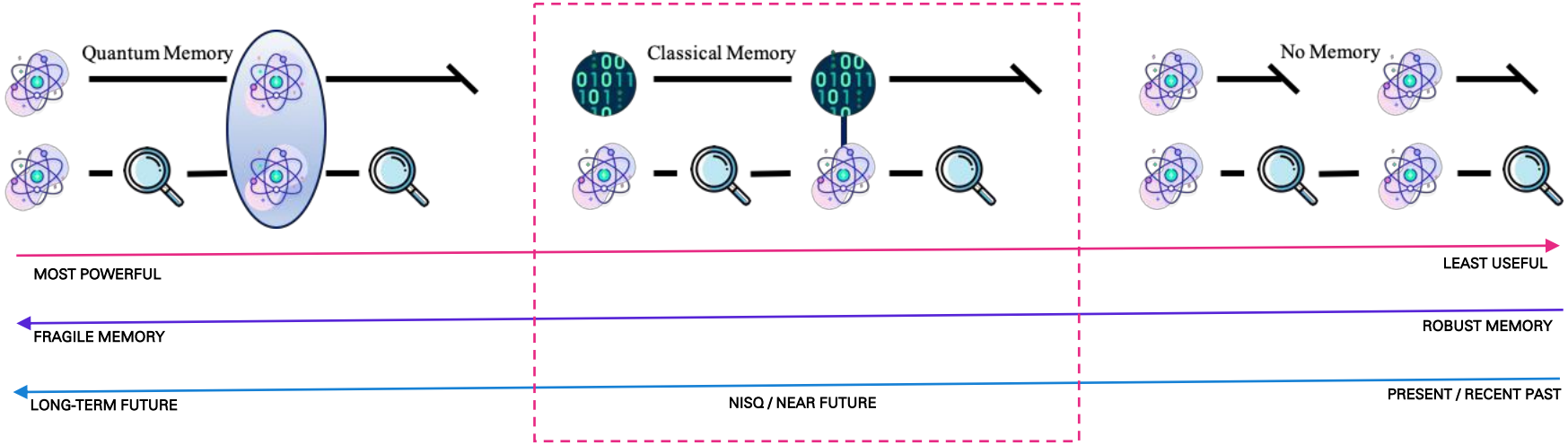


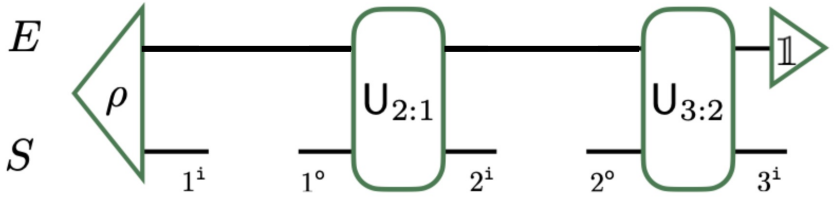
…yet at the same time can be controlled over significant timeframes **without being spoiled** by decoherence

However, the *distinction* between quantum and classical memory remains relatively *underexplored*

1. **Analyse** various physically motivated candidates regarding a **suitable definition** for classical memory
2. **Systematically characterise** the hierarchy of **multi-time memory effects** in quantum mechanics
3. Consider some **outlook** directions

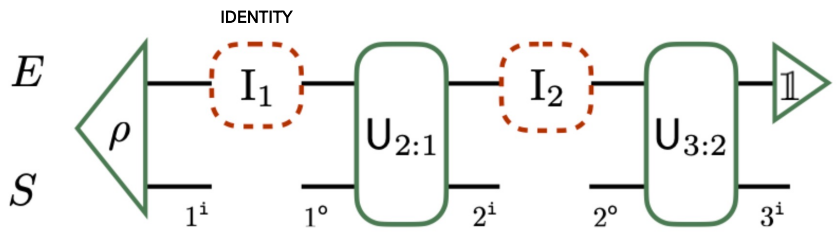
Multi-time Quantum Processes and Types of Memory

Framework: *Multi-time* Quantum Processes



Framework: *Multi-time* Quantum Processes

<u>Quantum Memory:</u> Environment remains coherent throughout evolution



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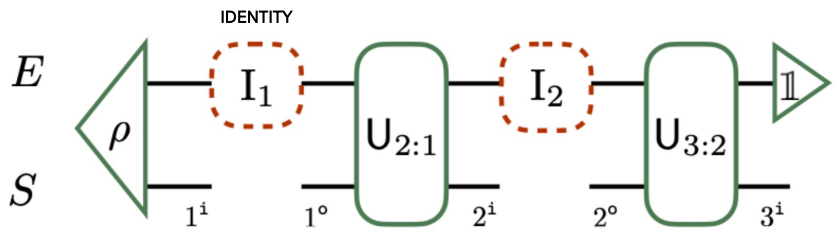
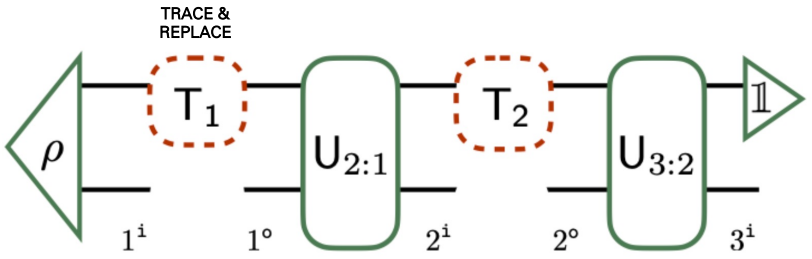


<u>Memoryless/Markovian Process:</u> Environment is discarded between every pair of times



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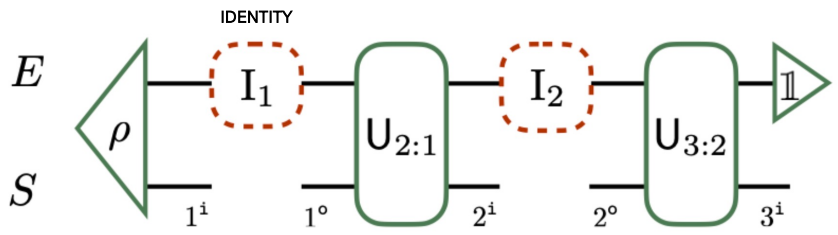

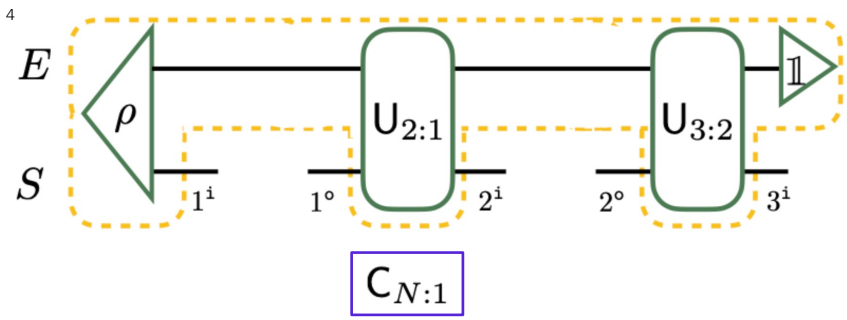

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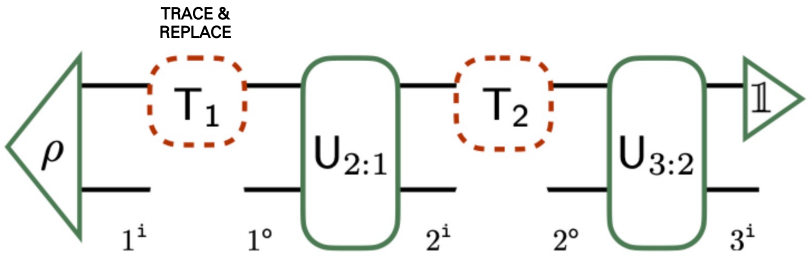

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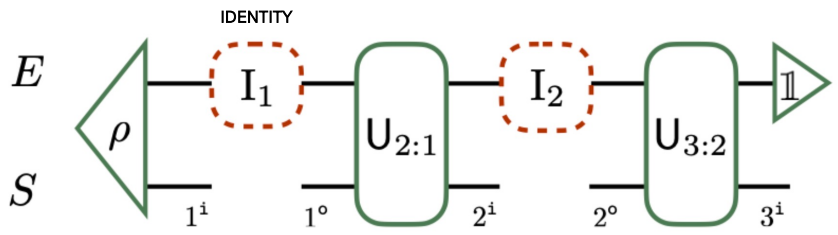

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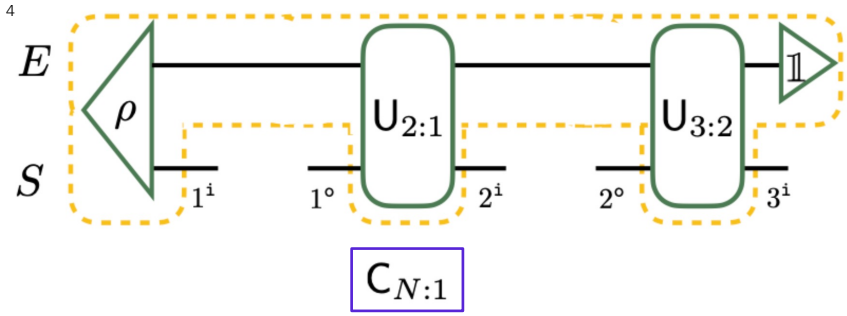

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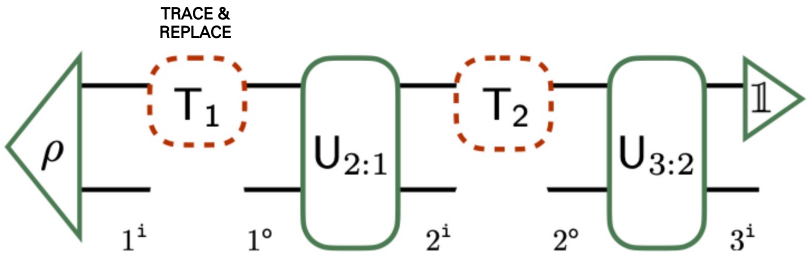

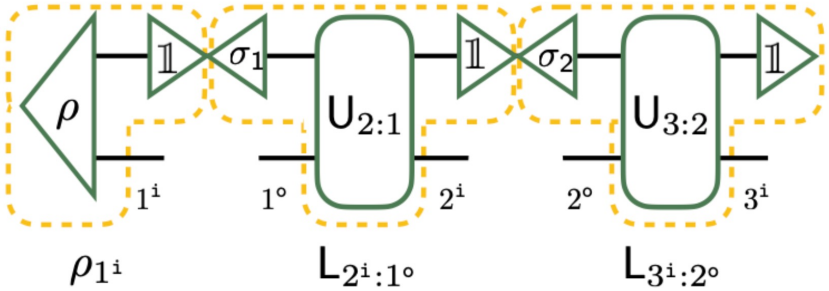

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[5] F. A. Pollock, C. Rodríguez-Rosario, T. Frauenheim, M. Paternostro, K. Modi, Phys. Rev. Lett. **120**, 040405 (2018); F. Costa, S. Shrapnel, New J. Phys. **18**, 063032 (2016)

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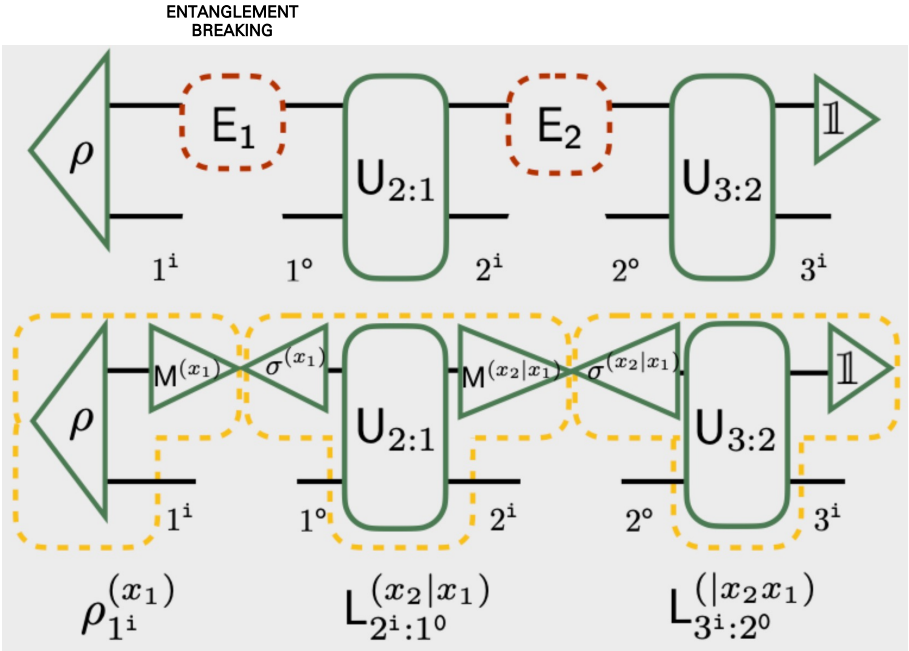
<u>Classical Memory:</u> Environment subject to entanglement-breaking channel between times



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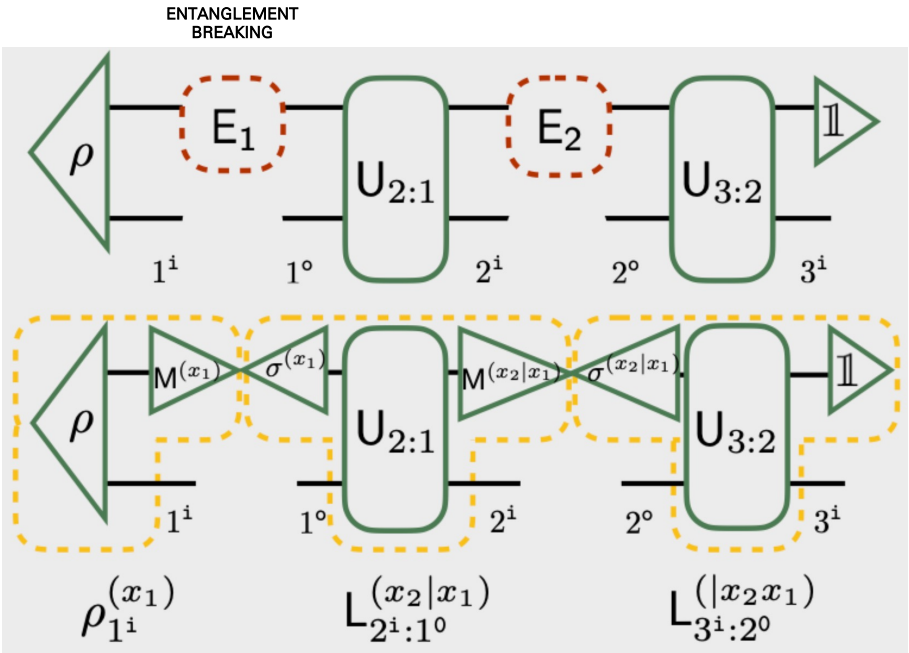

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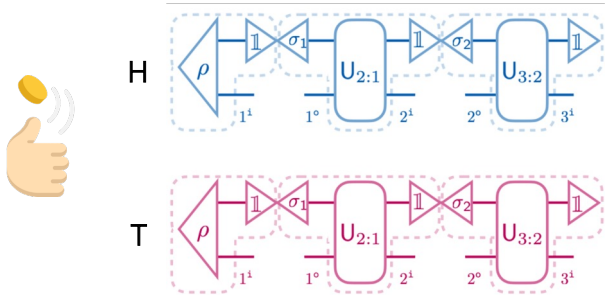
<u>Classical Direct Cause:</u>
Coin flip and randomly sample a memoryless process



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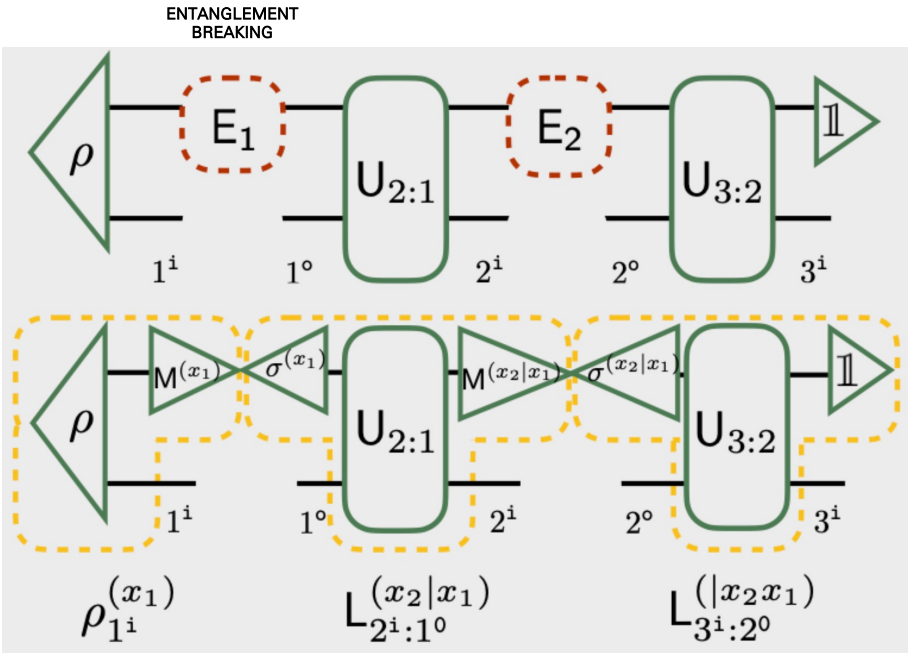

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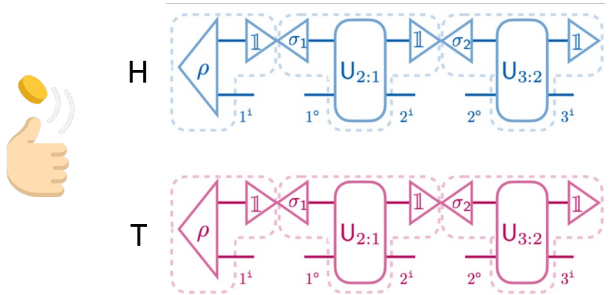

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<u>Separable Choi State:</u>
No "temporal entanglement" between times

$$\mathsf{C}_{N:1}^{\mathrm{SEP}} = \sum_{x} p_x \mathsf{L}_{N^{\mathrm{i}}:N-1^{\mathrm{o}}}^{(x)} \otimes \ldots \otimes \mathsf{L}_{2^{\mathrm{i}}:1^{\mathrm{o}}}^{(x)} \otimes \rho_{1^{\mathrm{i}}}^{(x)}$$

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Characterising the Hierarchy of Multi-time Quantum Processes

What is the *relationship* between these different classes of quantum processes?

Different types of memory

QUANTUM MEMORY

$$\mathsf{C}_{N:1}$$

CLASSICAL MEMORY

$$\mathsf{C}^{\mathrm{SEP}}_{N:1} = \sum_x p_x \mathsf{L}^{(x)}_{N^{\mathtt{i}}:N-1^{\mathtt{o}}} \otimes \ldots \otimes \mathsf{L}^{(x)}_{2^{\mathtt{i}}:1^{\mathtt{o}}} \otimes \rho^{(x)}_{1^{\mathtt{i}}}$$

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NO MEMORY

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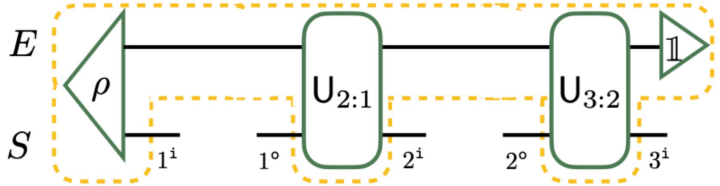

General non-Markovian quantum process / quantum comb

$\mathsf{C}_{N:1} \geq 0$ CP

$$\mathrm{tr}_{N^{\mathrm{i}}}[\mathsf{C}_{N:1}] = \mathbb{1}_{N-1^{\mathrm{o}}} \otimes \mathsf{C}_{N-1:1}$$
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What is the *relationship* between these different classes of quantum processes?

Different types of memory

QUANTUM MEMORY $\quad \mathsf{C}_{N:1}$

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$$\boxed{\mathsf{C}^{\mathrm{SEP}}_{N:1} = \sum_x p_x \mathsf{L}^{(x)}_{N^{\mathtt{i}}:N-1^{\mathtt{o}}} \otimes \ldots \otimes \mathsf{L}^{(x)}_{2^{\mathtt{i}}:1^{\mathtt{o}}} \otimes \rho^{(x)}_{1^{\mathtt{i}}}}$$

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$$\mathsf{C}^{\mathrm{CDC}}_{N:1} = \sum_x p_x \mathsf{L}^{(x)}_{N^{\mathtt{i}}:N-1^{\mathtt{o}}} \otimes \ldots \otimes \mathsf{L}^{(x)}_{2^{\mathtt{i}}:1^{\mathtt{o}}} \otimes \rho^{(x)}_{1^{\mathtt{i}}}$$

NO MEMORY $\quad \mathsf{C}^{\mathrm{M}}_{N:1} = \mathsf{L}_{N^{\mathtt{i}}:N-1^{\mathtt{o}}} \otimes \ldots \otimes \mathsf{L}_{2^{\mathtt{i}}:1^{\mathtt{o}}} \otimes \rho_{1^{\mathtt{i}}}$

Separable Choi State

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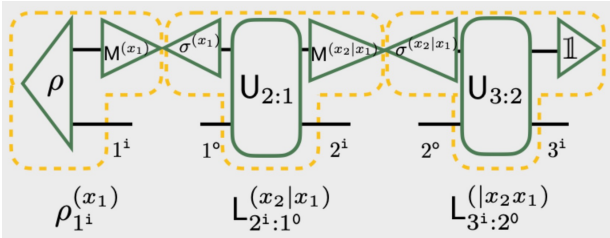

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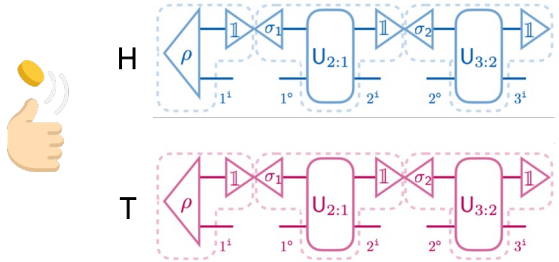

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$$\mathsf{C}^{\mathrm{CDC}}_{N:1} = \sum_x p_x \mathsf{L}^{(x)}_{N^{\mathtt{i}}:N-1^{\mathtt{o}}} \otimes \ldots \otimes \mathsf{L}^{(x)}_{2^{\mathtt{i}}:1^{\mathtt{o}}} \otimes \rho^{(x)}_{1^{\mathtt{i}}}$$

NO MEMORY

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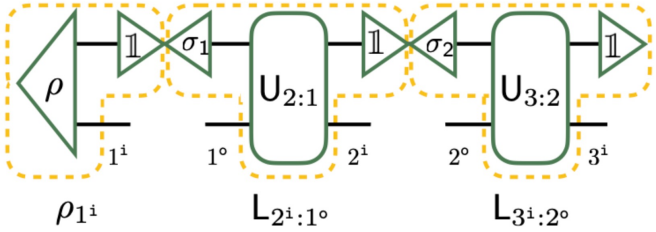

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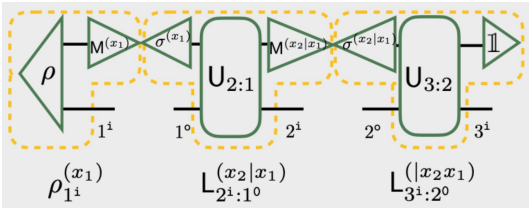

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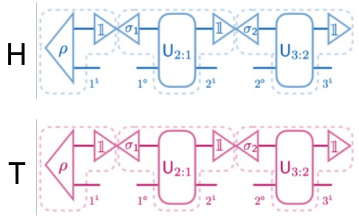

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EQUIVALENT FOR N=2

$$\boxed{\mathsf{C}_{N:1}^{\mathrm{CDC}} = \sum_x \mathsf{L}_{N^{\mathrm{i}}:N-1^{\mathrm{o}}}^{(x)} \otimes \ldots \otimes \mathsf{L}_{2^{\mathrm{i}}:1^{\mathrm{o}}}^{(x)} \otimes \rho_{1^{\mathrm{i}}}^{(x)}}$$

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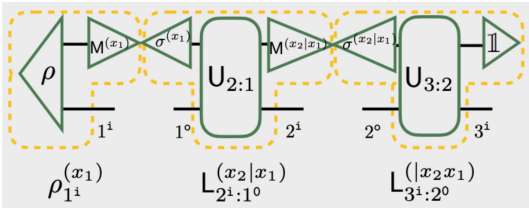

!= Separable Choi State

Different types of memory

QUANTUM MEMORY $\quad C_{N:1}$

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$$C_{N:1}^{\mathrm{SEP}} = \sum_x p_x L_{N^{\mathrm{i}}:N-1^{\mathrm{o}}}^{(x)} \otimes \ldots \otimes \rho_{1^{\mathrm{i}}}^{(x)}$$

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CONJECTURED INEQUIVALENCE IN [6]; PROVEN IN [7] (N=2)

$$C_{N:1}^{\mathrm{CDC}} = \sum_x p_x L_{N^{\mathrm{i}}:N-1^{\mathrm{o}}}^{(x)} \otimes \ldots \otimes L_{2^{\mathrm{i}}:1^{\mathrm{o}}}^{(x)} \otimes \rho_{1^{\mathrm{i}}}^{(x)}$$

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[6] C. Giarmatzi, F. Costa, Quantum **5**, 440 (2021)
[7] M. Nery, M. T. Quintino, P. A. Guérin, T. O. Maciel, R. O. Vianna, Quantum **5**, 538 (2021)

Characterising the hierarchy of memory effects

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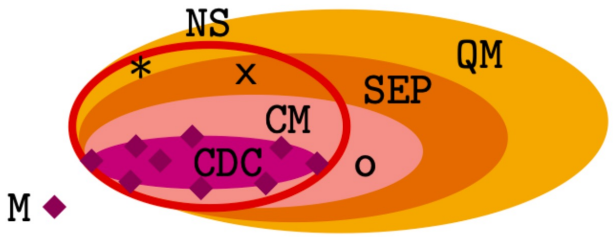
Demonstrate the strict hierarchy for N > 2:

$$\boxed{\mathrm{M} \subsetneq \mathrm{CDC} \subsetneq \mathrm{CM} \subsetneq \mathrm{SEP} \subsetneq \mathrm{QM}}$$



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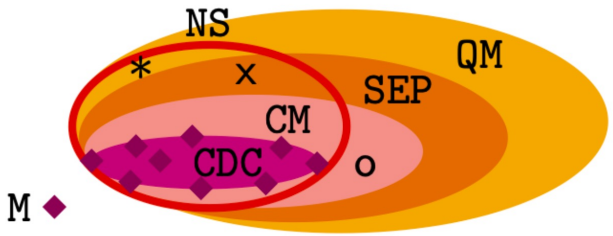
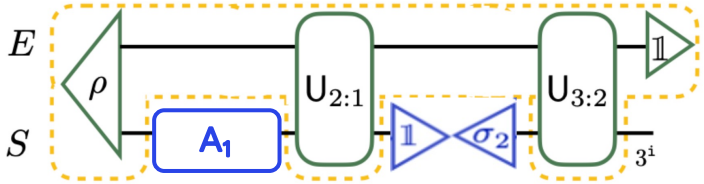


Introduce set of "non-signalling processes": such that if one discards the system state at any time, no historic information can be signalled to the future [7]



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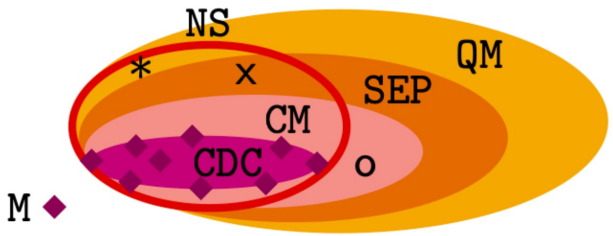

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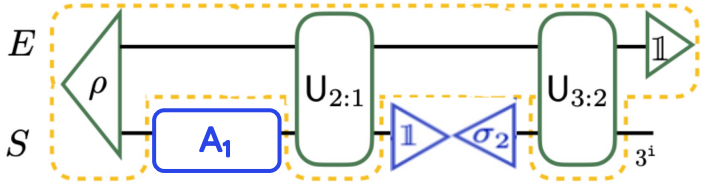

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Prove that there is no hierarchical relationship between CM, NS, and SEP in general… although there is for N=2:

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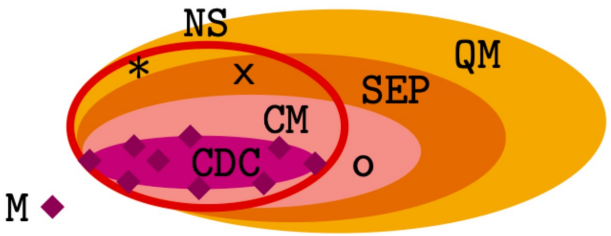

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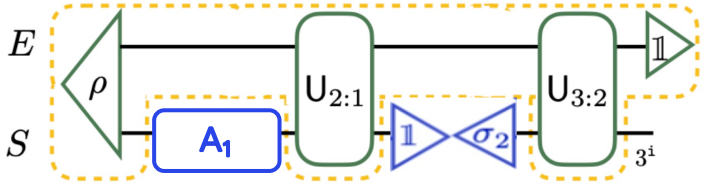

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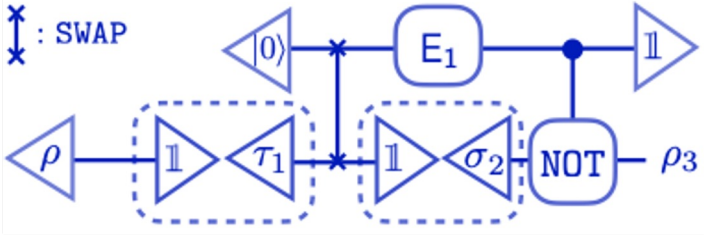
Examples:

o: CM but not NS (or therefore CDC)



x: SEP (and NS) but not CM [7]

$$\mathtt{C}^{\mathrm{SEP}}_{2:1} = \frac{1}{2}\left(|000\rangle\langle 000| + |101\rangle\langle 101| + |{+}1{+}\rangle\langle {+}1{+}| + |{-}1{-}\rangle\langle {-}1{-}|\right)$$

*: NS but not SEP

Any entangled state fed out over time.

$$\widetilde{\mathtt{C}}^{\mathrm{NS}}_{N:1} = \rho_{N^{\mathrm{i}}\cdots 1^{\mathrm{i}}} \otimes \mathbb{1}_{N-1^{\mathrm{o}}\cdots 1^{\mathrm{o}}}$$

[7] M. Nery, M. T. Quintino, P. A. Guérin, T. O. Maciel, R. O. Vianna, Quantum **5**, 538 (2021)

Outlook

Conclusions & future perspectives…

We have demonstrated that in the multi-time case, a **clear hierarchy** emerges amongst different types of memory (even though it collapses at the two-time level)

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Concretely connect these sets of processes to quantum information processing tasks for which each set is able to **achieve optimality**, as this would truly characterise them in terms of usefulness

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We show the distinctions by analysing the **signalling properties** of the processes, which link the potential memory effects to operationally implementable tests that can be determined by probing the system alone

In the paper, we also provide a **toolbox of computational methods**, some of which are efficiently implementable as SDPs, to numerically approximate each of the sets of processes considered

Concretely connect these sets of processes to quantum information processing tasks for which each set is able to **achieve optimality**, as this would truly characterise them in terms of usefulness

Uncover which kind of **continuous system-environment dynamics** lead to each different type of memory